

Lecture: Model Assessment and Selection

Actuarial Data Science - Open Learning Resource
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Recommended Reading

- The Elements of Statistical Learning (ESL): Chapter 7 (excluding Sections 7.8 and 7.9)
- An Introduction to Statistical Learning (ISLR): Review Sections 2.2, 4.4, 5, and 6.1.3



Learning Objectives

Up to this point, you have seen many modelling techniques. This lecture helps you answer a natural follow-up question: *which* model should I trust, and *how* do I know it will work on new data? We focus on practical tools for comparing models and understanding the trade-off between bias, variance, and complexity.

- Understand different measures and techniques for model assessment and selection.
- Explain bias, variance and model complexity.
- Use a range of statistical measures and techniques to assess model performance, select models, and distinguish between methods for regression and classification.
- Assess the appropriateness and performance of a model, taking into account business context and objectives.



Some Statistical Machine Learning Principles



No Free Lunch Theorem



No free lunch.

There is no single best model that works optimally for all kinds of problems.



Occam's Razor



The law of parsimony for problem-solving: “entities should not be multiplied without necessity.”

Bias, Variance and Model Complexity



Introduction

- The *generalisation* performance of a learning method relates to its prediction capability on independent test data.
- We introduce key methods for performance assessment and model selection.
- **Model selection:** estimating the performance of different models in order to choose the best one.
- **Model assessment:** having chosen a final model, estimating its prediction error (generalisation error) on new data.



Loss Function for Quantitative Response

- We have a *quantitative* response variable Y , a vector of inputs $\mathbf{X}$, and a prediction model $\hat{f}(\mathbf{X})$ estimated from a training set $\mathcal{T}$.
- The loss function for measuring prediction errors is $L(Y, \hat{f}(\mathbf{X}))$:

$$L(Y, \hat{f}(\mathbf{X})) = \begin{cases} (Y - \hat{f}(\mathbf{X}))^2 & \text{(squared error)} \\ |Y - \hat{f}(\mathbf{X})| & \text{(absolute error)} \end{cases}$$



Loss Function for Categorical Response

- The story is similar for a *categorical* response G taking one of K values ($1, 2, \dots, K$) in a set $\mathcal{G}$.
- Typically, we model the probabilities $p_k(\mathbf{X}) = \Pr(G = k \mid \mathbf{X})$.
- The predicted class is $\hat{G} = \arg \max_k \hat{p}_k(\mathbf{X})$.
- Loss functions:
 - $L(G, \hat{G}(\mathbf{X})) = I(G \neq \hat{G}(\mathbf{X}))$ (0-1 loss)
 - $L(G, \hat{p}(\mathbf{X})) = -2 \sum_{k=1}^K I(G = k) \log \hat{p}_k(\mathbf{X}) = -2 \log \hat{p}_G(\mathbf{X})$
 - $-2 \times \log$ -likelihood is sometimes referred to as the *deviance*
 - Can a loss function always align with your objective? (application-driven + convexity considerations)



Log-Likelihood Loss Function

- The log-likelihood can be used as a loss function for general response distributions, such as Poisson, Gamma, Exponential, log-normal, and others.
- If $\Pr_{\theta(\mathbf{X})}(Y)$ is the density of Y , indexed by a parameter $\theta(\mathbf{X})$ that depends on the predictor $\mathbf{X}$, then

$$L(Y, \theta(\mathbf{X})) = -2 \log \Pr_{\theta(\mathbf{X})}(Y).$$

- What if no likelihood is available?
- Why not use accuracy as the loss? (non-convex, NP-hard)



Errors for Quantitative Response

- **Test error** (generalisation error): the prediction error over an independent test sample

$$\text{Err}_{\mathcal{T}} = \mathbb{E}[L(Y, \hat{f}(\mathbf{X})) \mid \mathcal{T}]$$

- Both $\mathbf{X}$ and Y are drawn randomly from their joint distribution (population)
 - The training set $\mathcal{T}$ is fixed, and test error refers to the error for this specific training set
- **Expected test error** (expected prediction error): averages over all sources of randomness, including the randomness in the training set that produced $\hat{f}$

$$\text{Err} = \mathbb{E}[L(Y, \hat{f}(\mathbf{X}))] = \mathbb{E}[\text{Err}_{\mathcal{T}}]$$

- **Training error**: the average loss over the training sample

$$\overline{\text{err}} = \frac{1}{N} \sum_{i=1}^N L(y_i, \hat{f}(\mathbf{x}_i))$$



Errors for Categorical Response

- **Test error:** the population misclassification error of the classifier trained on $\mathcal{T}$

$$\text{Err}_{\mathcal{T}} = \mathbb{E}[L(G, \hat{G}(\mathbf{X})) \mid \mathcal{T}]$$

- **Expected test error** (expected misclassification error): the expected misclassification error

$$\text{Err} = \mathbb{E}[L(G, \hat{G}(\mathbf{X}))] = \mathbb{E}[\text{Err}_{\mathcal{T}}]$$

- **Training error** (example using log-loss / deviance):

$$\overline{\text{err}} = -\frac{2}{N} \sum_{i=1}^N \log \hat{p}_{g_i}(\mathbf{x}_i)$$



Model Assessment and Selection

- Assume the tuning parameter is α , and write the predictions as $\hat{f}_\alpha(x)$.
- The tuning parameter controls the complexity of the model.
- **Objectives:**
 - Find the value of α that minimises the *expected test error*
 - Assess the *generalisation error* of the final chosen model



Model Assessment and Selection (continued)

How can we do this?



- What if there is insufficient data to split into three parts? How much training data is enough?
- **Solutions:** approximate the validation step
 - Analytical approaches: AIC, BIC, MDL, SRM
 - Efficient sample re-use: cross-validation, bootstrap



The Bias-Variance Decomposition

- Model: $Y = f(\mathbf{X}) + \epsilon$, where $\mathbb{E}(\epsilon) = 0$ and $\text{Var}(\epsilon) = \sigma_\epsilon^2$.
- For an input point $\mathbf{X} = \mathbf{x}_0$, using squared-error loss, the *expected prediction error* is

$$\begin{aligned} \text{Err}(\mathbf{x}_0) &= \mathbb{E}[(Y - \hat{f}(\mathbf{x}_0))^2 \mid \mathbf{X} = \mathbf{x}_0] \\ &= \sigma_\epsilon^2 + (\mathbb{E}[\hat{f}(\mathbf{x}_0)] - f(\mathbf{x}_0))^2 + \mathbb{E}[(\hat{f}(\mathbf{x}_0) - \mathbb{E}[\hat{f}(\mathbf{x}_0)])^2] \\ &= \sigma_\epsilon^2 + \text{Bias}^2(\hat{f}(\mathbf{x}_0)) + \text{Var}(\hat{f}(\mathbf{x}_0)) \\ &= \text{Irreducible error} + \text{Bias}^2 + \text{Variance} \end{aligned}$$

- Typically, as model complexity increases, the (squared) bias decreases, but the variance increases.
- Examples: linear regression, k -nearest neighbour regression, ridge regression



Model Bias and Estimation Bias

- For ridge regression, let β_* denote the parameters of the best-fitting linear approximation to f :

$$\beta_* = \arg \min_{\beta} \mathbb{E}[(f(\mathbf{X}) - \mathbf{X}^\top \beta)^2]$$

- Assume the input variable $\mathbf{X}$ is random and the tuning parameter is α .
- For ridge regression, the average squared bias can be decomposed as:

$$\begin{aligned} \mathbb{E}_{\mathbf{x}_0} [f(\mathbf{x}_0) - \mathbb{E} \hat{f}_\alpha(\mathbf{x}_0)]^2 &= \mathbb{E}_{\mathbf{x}_0} [f(\mathbf{x}_0) - \mathbf{x}_0^\top \beta_*]^2 \\ &\quad + \mathbb{E}_{\mathbf{x}_0} [\mathbf{x}_0^\top \beta_* - \mathbb{E}(\mathbf{x}_0^\top \hat{\beta}_\alpha)]^2 \\ &= \text{Ave}[\text{Model bias}]^2 + \text{Ave}[\text{Estimation bias}]^2 \end{aligned}$$

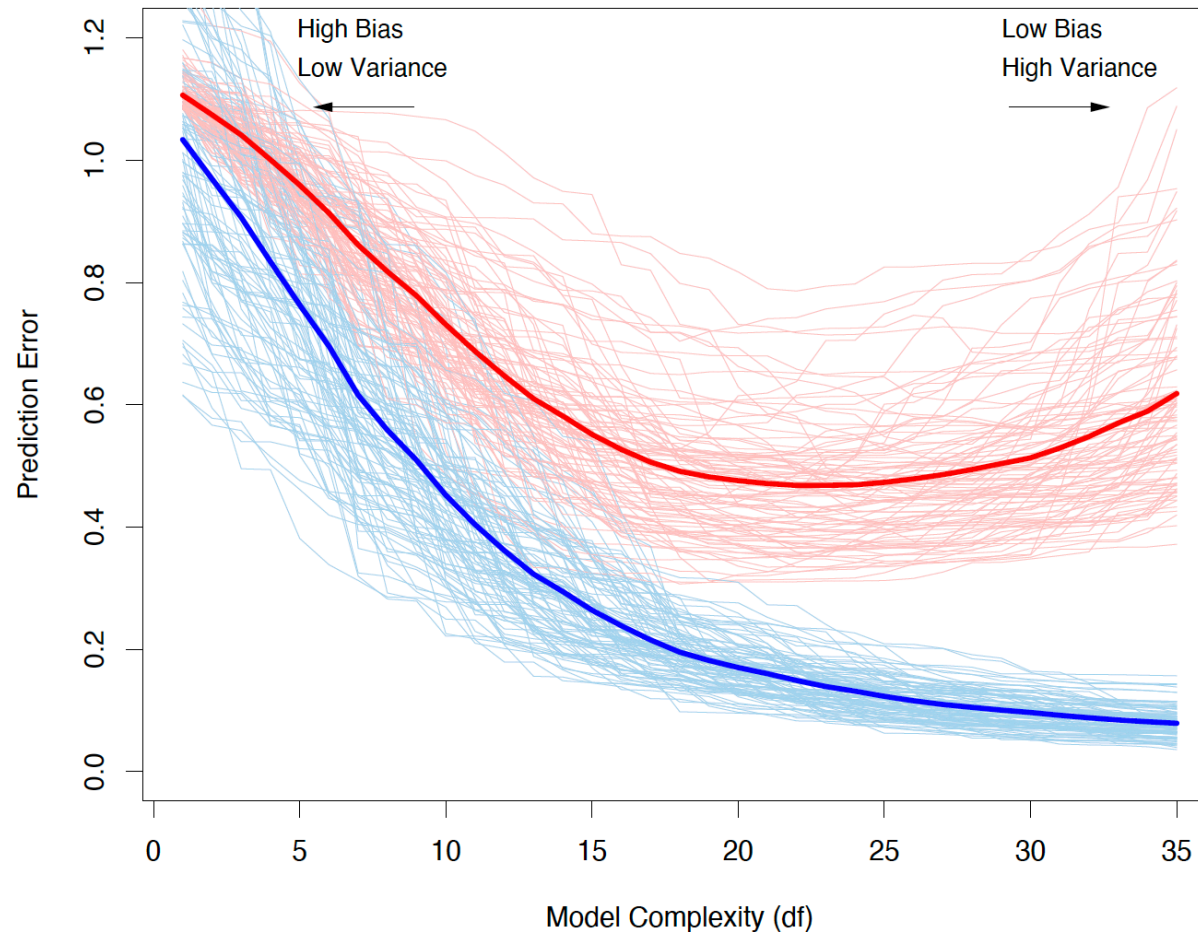


Model Bias and Estimation Bias (continued)

- **Model bias:** the error between the best-fitting linear approximation and the true function
 - How can we reduce model bias?
- **Estimation bias:** the error between the average estimate and the best-fitting linear approximation
 - OLS vs shrinkage



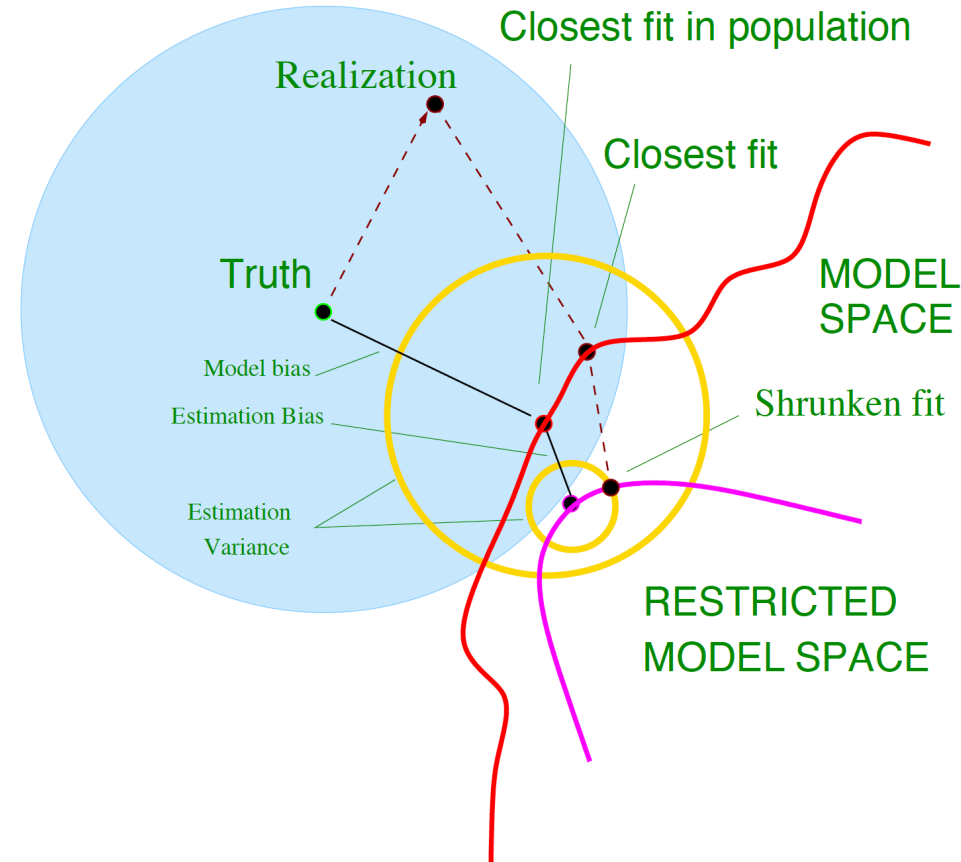
Bias, Variance and Model Complexity



Training and test error as a function of model complexity (Source: *The Elements of Statistical Learning* (Hastie et al. 2009), Figure 7.1).



Bias-Variance Trade-off



Schematic illustration of the bias-variance trade-off (Source: *The Elements of Statistical Learning* (Hastie et al. 2009), Figure 7.2).

Example

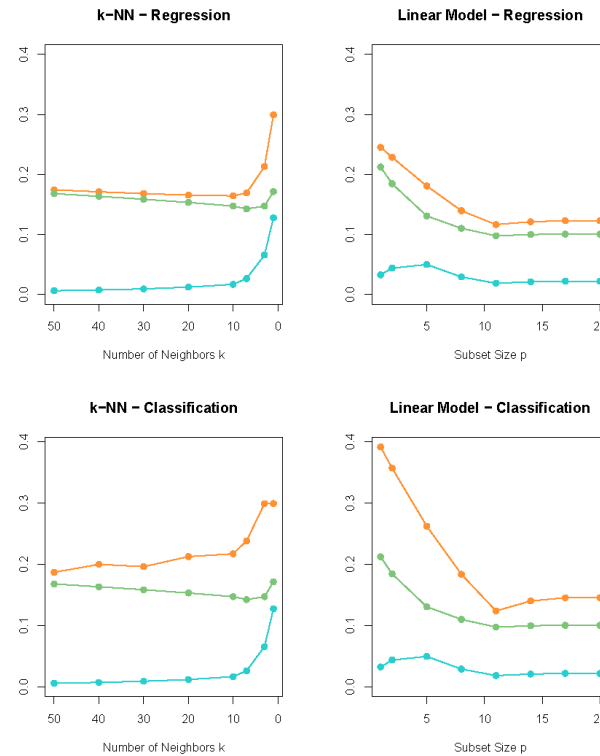


FIGURE 7.3. Expected prediction error (orange), squared bias (green) and variance (blue) for a simulated example. The top row is regression with squared error loss; the bottom row is classification with 0–1 loss. The models are k -nearest neighbors (left) and best subset regression of size p (right). The variance and bias curves are the same in regression and classification, but the prediction error curve is different.

Expected prediction error (orange), squared bias (green), and variance (blue) for k -NN (left) and linear models (right), for both regression (top) and classification (bottom) (Source: *The Elements of Statistical Learning* (Hastie et al. 2009), Figure 7.3).



Estimates of In-Sample Prediction Error



Optimism of the Training Error Rate

- *In-sample prediction error*

$$\text{Err}_{\text{in}} = \frac{1}{N} \sum_{i=1}^N \mathbb{E}_{Y^0} [L(Y_i^0, \hat{f}(\mathbf{x}_i)) \mid \mathcal{T}]$$

The notation Y^0 indicates that we observe N new response values at the training points $\mathbf{x}_i$, $i = 1, 2, \dots, N$.

- *Optimism* is the difference between Err_{in} and the training error $\overline{\text{err}}$: $\text{op} \equiv \text{Err}_{\text{in}} - \overline{\text{err}}$
- The *average optimism* is the expectation of the optimism over training sets: $w \equiv \mathbb{E}_{\mathbf{y}}(\text{op})$
- For squared error, 0–1 loss, and other loss functions, one can show quite generally that

$$w = \frac{2}{N} \sum_{i=1}^N \text{Cov}(\hat{y}_i, y_i)$$



In-sample Prediction Error

$$\mathbb{E}_{\mathbf{y}}(\text{Err}_{\text{in}}) = \mathbb{E}_{\mathbf{y}}(\overline{\text{err}}) + \frac{2}{N} \sum_{i=1}^N \text{Cov}(\hat{y}_i, y_i)$$

- For a linear model with d inputs (or basis functions), $Y = f(\mathbf{X}) + \epsilon$, we have $\sum_{i=1}^N \text{Cov}(\hat{y}_i, y_i) = d \sigma_{\epsilon}^2$, so

$$\mathbb{E}_{\mathbf{y}}(\text{Err}_{\text{in}}) = \mathbb{E}_{\mathbf{y}}(\overline{\text{err}}) + 2 \frac{d}{N} \sigma_{\epsilon}^2$$

- What happens as d increases or decreases?

Remarks:

- In-sample error is not usually of direct interest.
- In-sample error is convenient for model comparison and selection.
- The relative (rather than absolute) size of the error is what matters.



Estimates of In-sample Prediction Error

The general form of the in-sample estimate is

$$\widehat{\text{Err}}_{\text{in}} = \overline{\text{err}} + \hat{w}$$

where $\hat{w}$ is an estimate of the average optimism.

- $C_p = \overline{\text{err}} + 2 \frac{d}{N} \hat{\sigma}_\epsilon^2$

When a log-likelihood loss function is used:

- $\text{AIC} = -2 \log \text{lik} + 2d$
- $\text{BIC} = -2 \log \text{lik} + (\log N) d$

Another popular approach for model selection:

- Adjusted $R^2 = 1 - \frac{\text{RSS}/(n - d - 1)}{\text{TSS}/(n - 1)}$

Review: *An Introduction to Statistical Learning* (James et al. 2013), Section 6.1.3



Model Performance of Classification Models

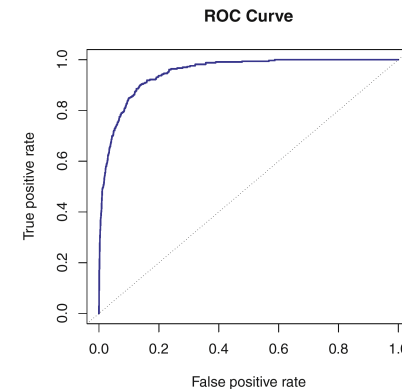
Performance Measures:

- AIC/BIC for logistic regression
- Accuracy = $\frac{TP + TN}{TP + FP + TN + FN}$
 - Not suitable for imbalanced class distributions
- Recall = $\frac{TP}{TP + FN}$
 - Important when the cost of false negatives is high (e.g. cancer screening)
- Precision = $\frac{TP}{TP + FP}$
 - Important when the cost of false positives is high (e.g. email spam detection)
- F1-score = $\frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$
 - Balances precision and recall
- ROC curve / Area under the ROC curve (AUC)

Review: *An Introduction to Statistical Learning* (James et al. 2013), Section 4.4

		Actual Values	
		Positive (1)	Negative (0)
Predicted Values	Positive (1)	TP	FP
	Negative (0)	FN	TN

Confusion matrix.



ROC curve (Source: *An Introduction to Statistical Learning* (James et al. 2013), Figure 4.8).



Estimates of Extra-sample Prediction Error

Review: *An Introduction to Statistical Learning* (James et al. 2013), Chapter 5



Different Scenarios

1. **Scenario 1:** Train a simple model (no hyperparameters)

- Training + Testing

2. **Scenario 2:** Train a model and tune its hyperparameters

- *Validation set approach or cross-validation*

3. **Scenario 3:** Compare different models (with hyperparameters), e.g. Random Forest vs Neural Network

- *nested cross validation*

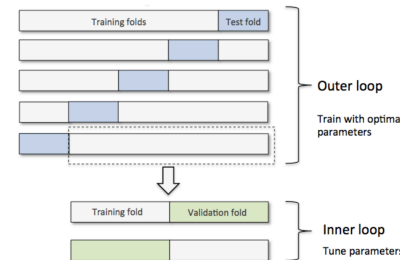


Illustration of nested cross-validation (Source: Raschka (2018)).

- The appropriate approach depends on the problem and the available data.



Cross-Validation (CV)

- Resampling without replacement
- CV directly estimates $\text{Err} = \mathbb{E}[L(Y, \hat{f}(X))]$, the average generalisation error
- Avoid information leakage
 - CV must be applied to the entire sequence of modelling steps
 - Samples must be “left out” before any selection or filtering steps are applied
 - Initial *unsupervised* screening steps can be performed before samples are left out
- Underlying assumption:
 - Training and test sets are drawn from the same population
- What value should we choose for K ?
 - Training-set bias: leave-one-out CV has low bias but high variance
 - In practice, 5- or 10-fold CV is often a good compromise
 - The choice depends on the objective



Bootstrap

- Resampling with replacement
- As with cross-validation, the bootstrap seeks to estimate the conditional error $\text{Err}_{\mathcal{T}}$, but typically performs well only for estimating the expected prediction error Err .
- Quantifying uncertainty:
 - See [Raschka \(2018\)](#)
- Implementation:
 - [R package bootstrap](#)



Bootstrap (continued)

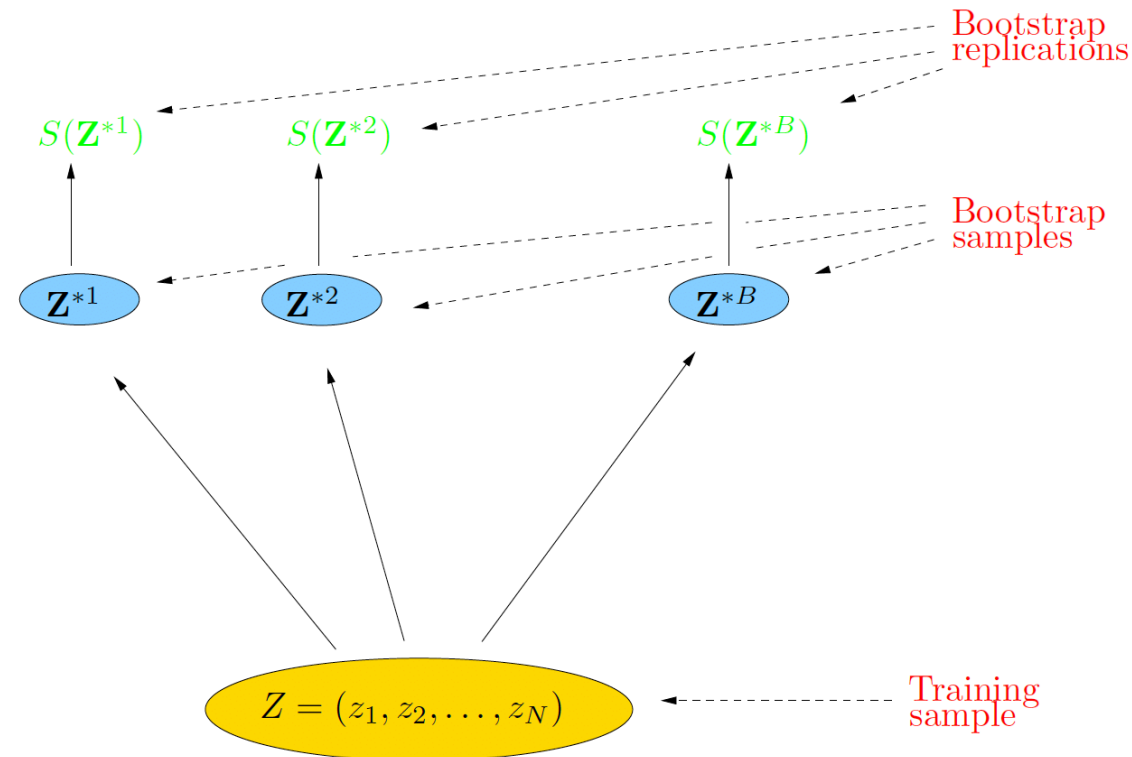


FIGURE 7.12. Schematic of the bootstrap process. We wish to assess the statistical accuracy of a quantity $S(\mathbf{Z})$ computed from our dataset. B training sets $\mathbf{Z}^{*b}$, $b = 1, \dots, B$ each of size N are drawn with replacement from the original dataset. The quantity of interest $S(\mathbf{Z})$ is computed from each bootstrap training set, and the values $S(\mathbf{Z}^{*1}), \dots, S(\mathbf{Z}^{*B})$ are used to assess the statistical accuracy of $S(\mathbf{Z})$.

Bootstrap sampling illustration (Source: *The Elements of Statistical Learning* (Hastie et al. 2009), Figure 7.12).

Bootstrap Estimators I

- If $\hat{f}^{*b}$ is the predicted value at $\mathbf{x}$ from the model fitted to the b -th bootstrap sample, an estimate of the expected prediction error is

$$\widehat{\text{Err}}_{\text{boot}} = \frac{1}{B} \frac{1}{N} \sum_{b=1}^B \sum_{i=1}^N L(y_i, \hat{f}^{*b}(\mathbf{x}_i)).$$

- Is this a good estimator? Why?



Bootstrap Estimators I (continued)

- Alternatively, for each observation, we only use predictions from bootstrap samples that do not contain that observation. The *leave-one-out bootstrap* estimate of prediction error is

$$\widehat{\text{Err}}^{(1)} = \frac{1}{N} \sum_{i=1}^N \frac{1}{|C^{-i}|} \sum_{b \in C^{-i}} L(y_i, \hat{f}^{*b}(\mathbf{x}_i)).$$

- Here C^{-i} is the set of bootstrap samples that do not contain observation i , and $|C^{-i}|$ is the number of such samples
- Training-set-size bias: $\Pr\{i \in \text{bootstrap sample}\} = 1 - (1 - 1/N)^N \approx 1 - e^{-1} \approx 0.632$



Bootstrap Estimators II

- The “.632” estimator:

$$\widehat{\text{Err}}^{(.632)} = 0.368 \overline{\text{err}} + 0.632 \widehat{\text{Err}}^{(1)}$$

- Works well in *light overfitting* situations, but can break down under severe overfitting
 - Can be improved by accounting for the degree of overfitting
- The “.632+” estimator:

$$\widehat{\text{Err}}^{(.632+)} = (1 - \hat{w}) \overline{\text{err}} + \hat{w} \widehat{\text{Err}}^{(1)}$$

- $\hat{w} = \frac{0.632}{1 - 0.368 \hat{R}} \in [0.632, 1]$
 - The *relative overfitting rate*: $\hat{R} = \frac{\widehat{\text{Err}}^{(1)} - \overline{\text{err}}}{\hat{\gamma} - \overline{\text{err}}} \in [0, 1]$
 - The *no-information error rate*: $\gamma = \frac{1}{N^2} \sum_{i=1}^N \sum_{i'=1}^N L(y_i, \hat{f}(\mathbf{x}_{i'}))$



Model Assessment and Selection in a Business Context



Model Assessment and Selection in a Business Context

- Popular algorithms evolve rapidly
 - Top 10 algorithms in data mining in 2007 (Wu et al. 2008)
- Model assessment and selection should take into account business context and objectives
- Key questions:
 - Which loss function should be used?
 - Which performance measure is most suitable?
 - Which model selection method should be applied?



References

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